# Lab Report - 3

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November 2024

## 1 Abstract

The objective of this experiment is to understand the interaction of particles with matter which is particularly related to dose calculations for radiotherapy. This was done using the simulation software GEANT4 which provides three types of files: Dose profile, Longitudinal dose profile and Transverse dose profile. Three beam types were used: Photon, Proton and Carbon ions. The energy values were varied to understand the depth the beam could reach the phantom and the corresponding magnitudes. Simulations were done with the Tumor and with Hadronic interactions. The Tumor to Normal ratio (TNR) is evaluated at different energy values.

## 2 Background and Introduction

Radiotherapy is where a beam of high energy subatomic particles are used to cause cell damage to unhealthy tissues. This damage is done at a DNA level via microscopic interactions where the beam particles transfers energy to the tissue which results in causing the bonds to break and damages the unhealthy cell tissue. In order to make sure that the healthy tissues have very minimum impact due to the energy beams the beams can be projected at different orientations targeting the unhealthy area. Simulation used is GEANT4 which is a toolkit designed to create simulations of radiation through matter. Physical processes incorporated in the simulation are electromagnetic, strong and weak interactions of particles.
At the beginning the beam types at different energies are being evaluated to understand its behavior. Afterwards when the tumor is on it the simulations were repeated to observe the maximum TNR value which would mean that the damage caused in the patient's healthy tissue is minimized. Hadronic interactions are introduced as it is utilized to precisely deliver radiation to the tumor minimizing the damage to the healthy tissues by using the properties of the Bragg peak where energy is located at the end of the particles path concentrating the dose with the tumor volume. An intense analysis of the beams would give an idea of the appropriate beam and beam energy that could be used reaching the target with less damage.
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## 3 Theory

The primary focus of physics in Radiotherapy is to increase the accuracy of the dose delivery to the target area and minimize the impact it can cause on the healthy tissues. Therefore, there are many inventions conducted such as technological inventions in dose delivery and imaging. Ionizing radiation can cause serious impact to other surrounding tissues, therefore identifying the lethal dose is of high importance. This is generally accomplished by projecting the beam to the affected area through multiple pathways that cross the tumor region to avoid the overexposure of the healthy tissues. Another technique utilized is projecting at small doses over several sessions as then the healthy tissues have time to repair. Intensity Modulation Radiation Therapy (IMRT) is a highly recognized technique that is uses high energy proton beams for dose delivery. [1] Charged particle beam therapy is where the beams of protons, carbon ions are used instead of photons. Charged particles in comparison to photon beams have different depth dose distributions. More energy is deposited in the last stages of the trajectory resulting in a sharp, localized dose, defined as the Bragg peak. The initial energy of the beam determines the depth in the body the Bragg peak will form. Intensity of the beam determines the the magnitude of the dose that deposits in the body. Adjusting the beam energy it is possible to vary the dose with high precision. When the radiation dose increases the biological effects of

the charged particles increase. Charged particles have the possibility of covering the target area in lateral dimensions and depth with a single radiation beam which cannot be seen when using photon beams. Charged particles are able to minimize the dose reduction in healthy tissues in comparison to photon therapy.
The cell damage caused due to the exposure of ionizing radiation is identified as the Equivalent dose (H). The equivalent dose absorbed by a tissue can be defined as,

$$H = \Sigma_R W_R D_R$$

where $W_R$: weighting factor for a radiation type R which measures the relative biological effectiveness of the radiation type (RBE). $D_R$ is the absorbed dose which is mean energy deposited due to ionizing radiation in matter per mass unit.

$$D = \frac{E_d}{m}$$

Electromagnetic energy loss due to the heavy charged particles as a result of the ionizing process is described using the Bethe-Bloch. When the particle speed decreases the interactions with atomic electrons increases which leads to higher energy loss until the particle comes to rest. This Bragg peak characteristic is used in proton and ion therapy. The Tumor to Normal ration (TNR) is,

$$TNR = \frac{D_{tumor}}{D_{healthy}}$$

The main objective is to make sure the TNR value is at its maximum in order to reduce the damage cause to the surrounding healthy tissues.
Hadronic interactions uses hadrons in radiotherapy where they release energy differently to photons. They release energy slowly until they reach the critical speed and then rapidly decrease and releases a large amount of energy in a narrow area known as the Bragg peak. These interactions reaches the target area causing very minimum impact to the surrounding area. They travel parallel to the body and the Bragg peak is concentrated at the tumor region. Different types of particles are used in Hadron-therapy such as protons and heavy charged particles.
[Word Count: 543]

# 4 Experimental Methods

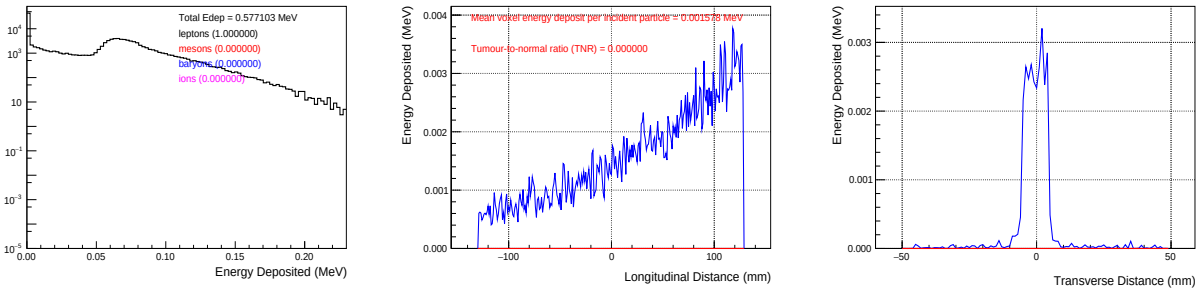
The simulation software used in this experiment is GEANT4. Three beams was used in the evaluation: photon beams, proton beams and carbon ion beams. Photon beams were at energies 1-50 MeV, Proton beams were at 100-100 MeV and Carbon ion beams were at 500-2000 MeV. The simulation was executed using the UNIX commands in the environment in which GEANT4 is installed. **./runsim.sh [beamtype] [energy in MeV]** For an instance to run the proton beam at 1000 MeV the command line should be as ./runsim.sh proton 1000. There are three simulation outputs:

*G4_proton_1000P.pdf* : dose profile by particle type
*G4_proton_1000L.pdf* : Longitudinal dose profile
*G4_proton_1000T.pdf* : Transverse dose profile

These dose profile files show the calculated total energy deposited in the phantom per incident per particle (MeV). The histogram depicts the frequency distribution of the deposited energy for different particle types. These particles can be leptons, mesons, baryons or ions. The longitudinal and transverse files exhibits the distribution of energy deposit in phantom in longitudinal: (parallel to the beam) and transverse: (perpendicular to the beam) directions.
At the begining the three beams were evaluated at different energies and obtained the P,L, and T files. Next was to turn on the tumor which was done by editing the first line of macros/DetectorSetup.mac: /DoseProfileMC/detector/SetTumorOn 1. Initially it was at 0 and to turn on the tumor it needs to be changed to 1. The objective is to obtain the maximum TNR value for different energies. This included trying out different energies by doing a binary search to figure out the maximum value. It was observed that for gamma beams the TNR value is at a lower energy and for carbon ions it is at a considerably higher value. Once the energies corresponding to the maximum TNR values are obtained, the next step was to turn on Hadronic interactions. This was done by editing the line in the macros/batch.mac.file and changing /DoseProfileMc/physics/addPhysics standard_opt3 to /DoseProfileMc/physics/addPhysics QGSP_BIC_EMY. The error analysis were conducted by taking the difference between the TNR values to extremely small changes in the energy. It was observed that the high energy carbon beams and proton beams efficiently reached the target area with a high TNR value in comparison to photon beams. This is due to the fact that photon beams expose radiation to healthy regions as well, causing cell damage whereas proton beams project radiation to a more concentrated area.
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# 5 Analysis and Results

**Question 1:**

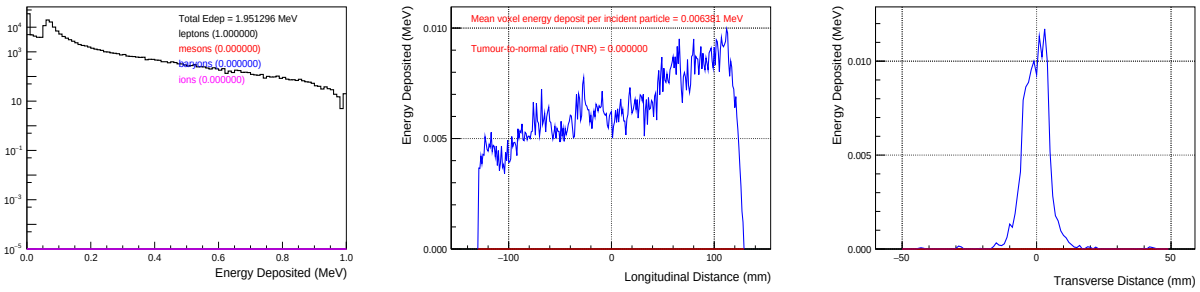



(a) Dose Profile (b) Longitudinal Dose Profile (c) Transverse Dose Profile

Figure 1: Gamma beam: 1 MeV

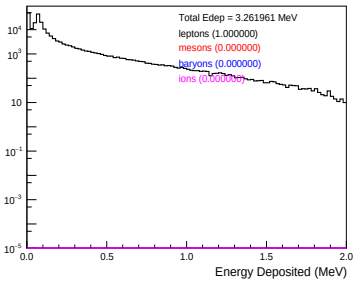
Figure 1 denotes the three output files for the photon beam at 1 MeV. Here it can be seen that the deposited energy is 0.577 MeV and the lepton contribution is 100%. When photon beams interact with matter the secondary particles created are leptons which are responsible for the majority tissue damage and occurs due Compton scattering and the Photoelectric effect. The longitudinal dose profile exhibits the energy deposition when the beam is traveling parallel to the phantom. The mean voxel energy is around 0.00158 MeV. The transverse dose profile shows the energy deposition when the particle beam is traveling perpendicular to the phantom. It can be observed that the photons looses energy as it passes through matter. Therefore it distributes the energy to the surrounding tissues causing cell damage to the healthy tissues.

**Question 2:**

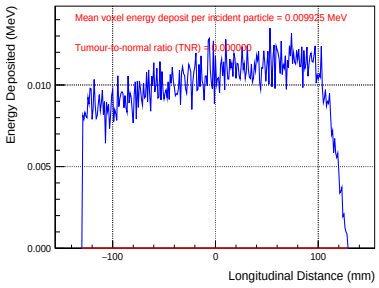



(a) Dose Profile (b) Longitudinal Dose Profile (c) Transverse Dose Profile

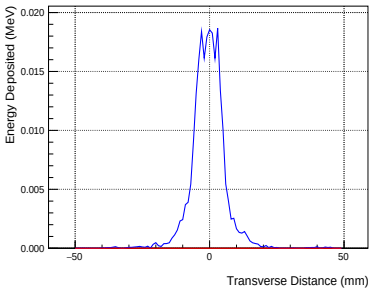
Figure 2: Gamma beam: 5 MeV




(a) Dose Profile

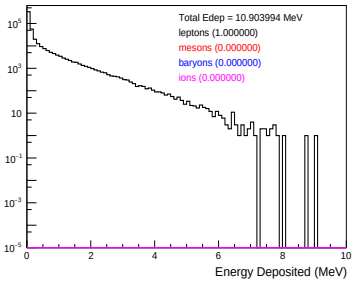



(b) Longitudinal Dose Profile

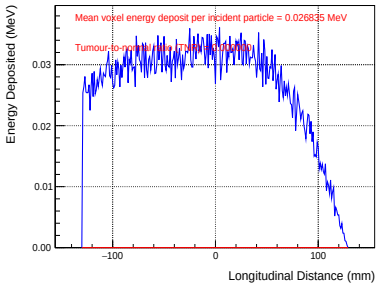



(c) Transverse Dose Profile

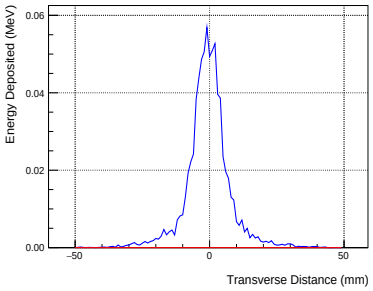
Figure 3: Gamma beam: 10 MeV




(a) Dose Profile




(b) Longitudinal Dose Profile

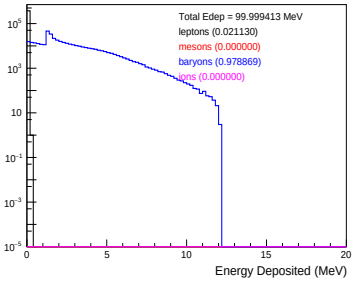



(c) Transverse Dose Profile

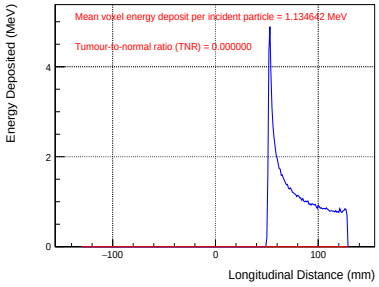
Figure 4: Gamma beam: 50 MeV

Energy deposition for the photon beams at incident energy 5 MeV, 10 MeV and 50 MeV are 1.951 MeV, 3.262 MeV and 10.904 MeV respectively. The deposition energy increases as the incident energy increases. It can be observed that as the incident energy increases the longitudinal dose profile reaches a higher mean voxel energy and immediately drops. The beam is at a high consistent energy value throughout the phantom damaging the rest of the tissues as well. The transverse dose profile shows that the radiation distributed perpendicular to the phantom reaches a higher value at the center point showing an equal distribution of energy throughout the region.

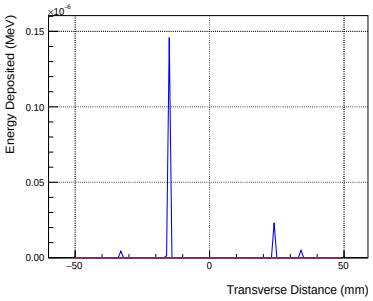
**Question 3:**




(a) Dose Profile

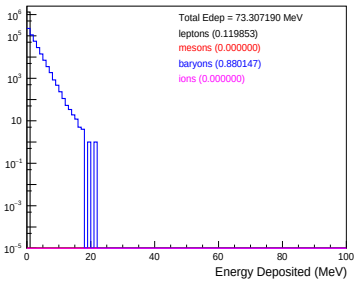



(b) Longitudinal Dose Profile

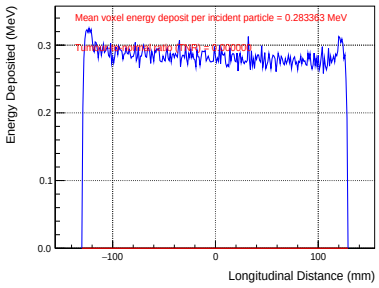



(c) Transverse Dose Profile

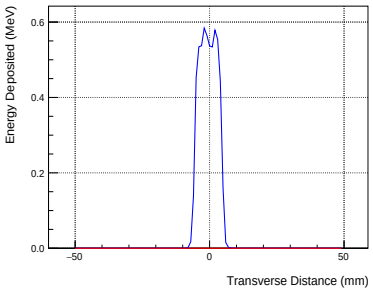
Figure 5: Proton beam: 100 MeV




(a) Dose Profile

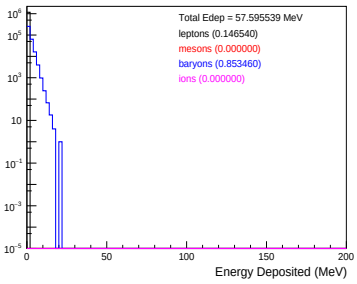



(b) Longitudinal Dose Profile

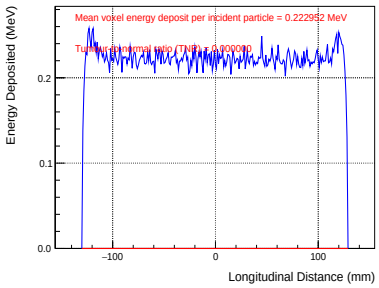



(c) Transverse Dose Profile

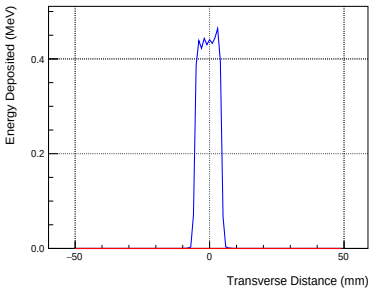
Figure 6: Proton beam: 500 MeV




(a) Dose Profile



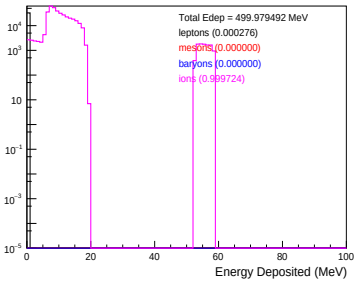

(b) Longitudinal Dose Profile



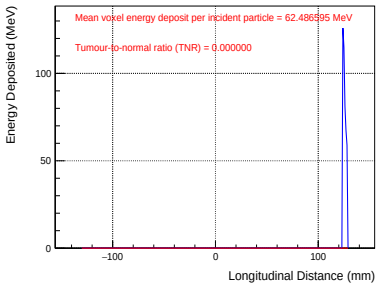

(c) Transverse Dose Profile

Figure 7: Proton beam: 1000 MeV

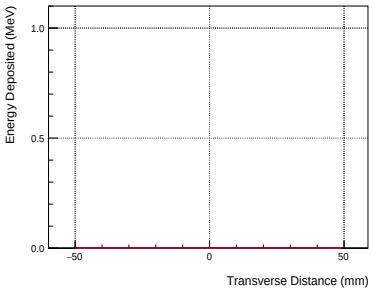
The figures 5,6,7 demonstrates the energy output files for the proton beam at energies 100 MeV, 500 MeV and 1000 MeV respectively. Secondary particles observed are leptons and baryons. In the longitudinal dose profiles the Bragg peaks can be observed when the proton beam is at 100 MeV. The FWMH is steep at this energy level and the mean voxel energy deposit per particle is 1.1347 MeV which is relative higher than the other energies. A sharp increase of energy can be seen at about 40-50 mm longitudinal distance and a sharp decrease of the energy can be seen at about 120 mm longitudinal distance. This means that the energy deposition is concentrated at the tumor region. The distribution of energy in the transverse dose profile shows a sharp peak and couple minor peaks but it does not fully reflect the accurate energy distribution. When the energy is increased in the photon beams the energy distribution is high and it affects the entire region of the tumor damaging the healthy cells. Therefore, 500 MeV and 1000 MeV proton beams should not be used for proton therapy.




(a) Dose Profile

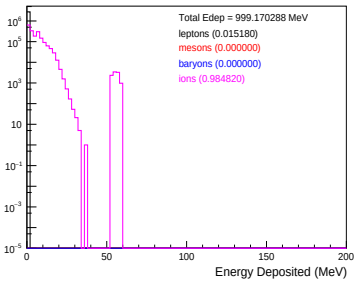



(b) Longitudinal Dose Profile

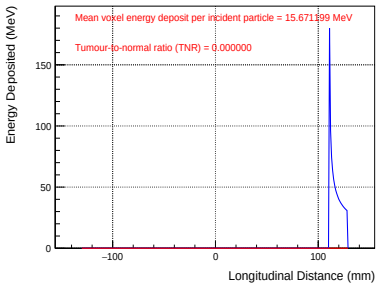



(c) Transverse Dose Profile

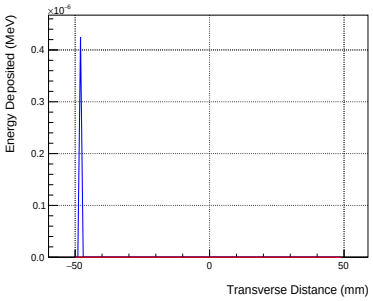
Figure 8: Carbon ion beam: 500 MeV




(a) Dose Profile

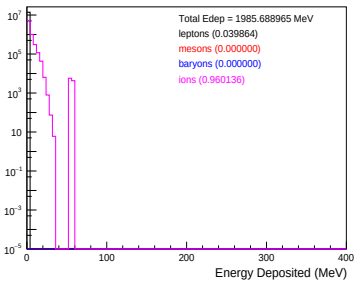



(b) Longitudinal Dose Profile

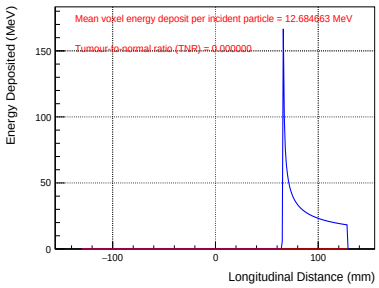



(c) Transverse Dose Profile

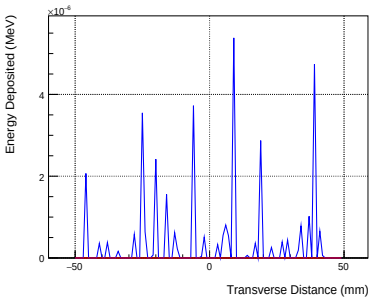
Figure 9: Carbon ion beam: 1000 MeV




(a) Dose Profile




(b) Longitudinal Dose Profile




(c) Transverse Dose Profile

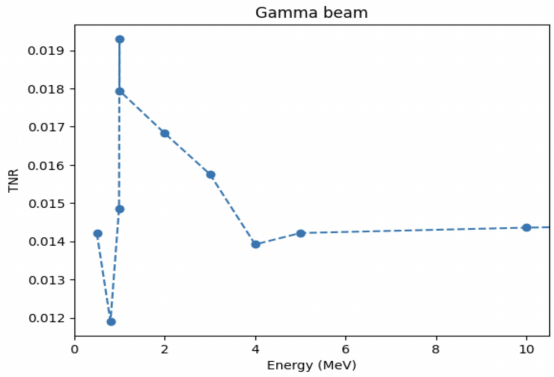
Figure 10: Carbon ion beam: 2000 MeV

The carbon ion beams concentrate the energy deposition in the target area more precisely comparative to photon and proton beams as can be observed in Figures 8,9 ans 10. Mean voxel energy deposit per incident particle is high at 500 MeV which accounts to 62.487 MeV but the total energy deposited is lower. The energy is concentrated to a small volume. The secondary particles formed are leptons and ions. The totally energy deposited is higher than the protons and photons. In carbon ions as the incident energy increases the longitudinal distance spectrum in which the energy deposits widens. The FWMH is much steeper in carbon ions that in the proton beam. It can be said that carbon ion beams are better for radiation therapy as a higher linear energy transfer which reflects that more radiation is projected to the tumor than the surrounding healthy tissues.

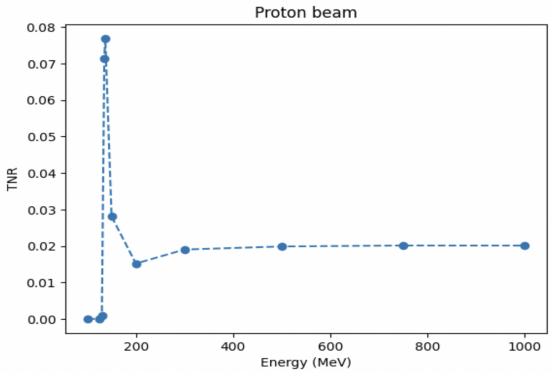
**Question 4:**
The second aspect of the experiment was to turn the tumor on and identify the energy in which the TNR is at it's maximum value. Figure 11 shows the TNR values and the corresponding energies. Gamma beam reaches a higher TNR value at a extremely small energy value whereas the Proton beam reaches at a higher energy than the photon beam. The variation of the Carbon ion beam in Figure 11(c) shows that the TNR value is at 0 until it reaches a higher energy level approximately closer to 3000 MeV. It can be seen in Figures 12, 13 and 14 that the Bragg peak is at the center of the tumor and sharp energy is concentrated at the target.

Table 1: Maximized TNR values for respective energies

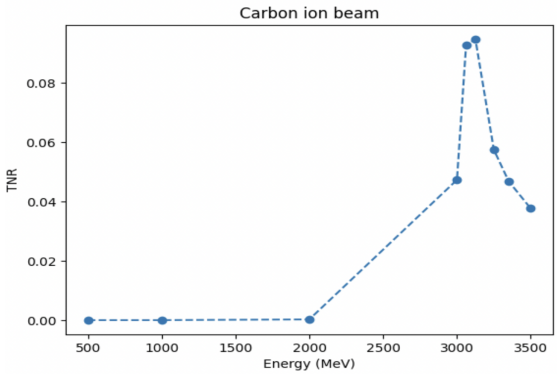
| **Beam** | **Energy (MeV)** | **TNR value** | **Error in TNR** |
|---|---|---|---|
| Gamma beam | 0.999 | 0.019300 | 0.0014 |
| Proton beam | 137 | 0.076894 | 0.0488 |
| Carbon ion beam | 3125 | 0.094831 | 0.0019 |




(a) Gamma beam

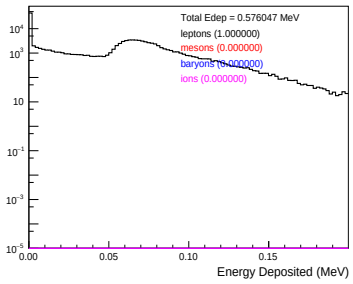



(b) Proton beam

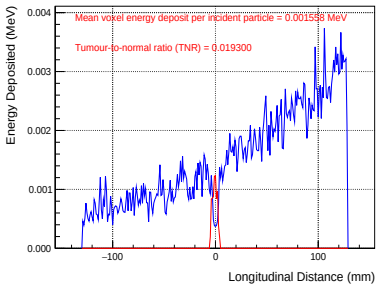



(c) Carbon ion beam

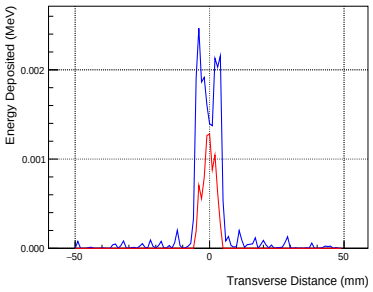
Figure 11: TNR vs Energy (MeV) plot




(a) Dose Profile

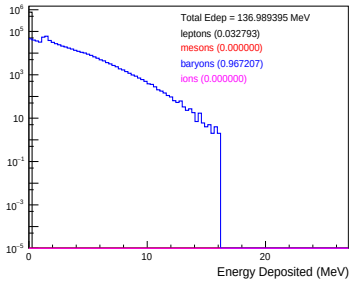



(b) Longitudinal Dose Profile

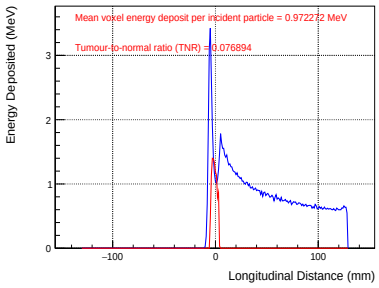



(c) Transverse Dose Profile

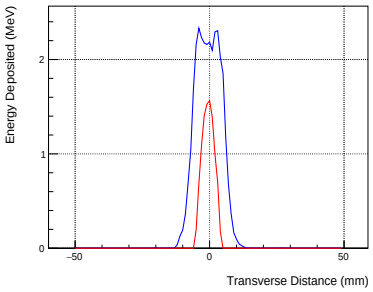
Figure 12: Gamma beam: 0.999 MeV




(a) Dose Profile

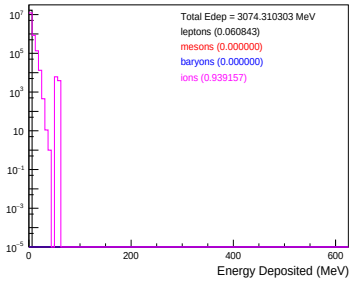



(b) Longitudinal Dose Profile

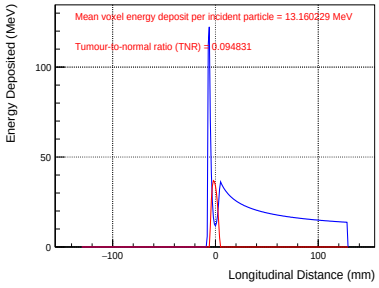



(c) Transverse Dose Profile

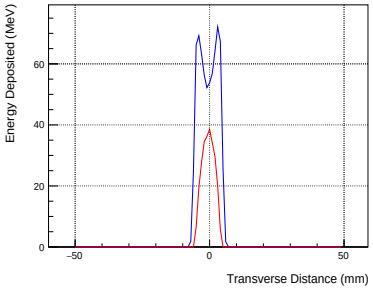
Figure 13: Proton beam: 137 MeV




(a) Dose Profile




(b) Longitudinal Dose Profile




(c) Transverse Dose Profile

Figure 14: Carbon ion beam: 3125 MeV

**Question 5:**
The final aspect of the experiment was to evaluate the variations when Hadronic interactions are introduced. A basic observation would be that the energy deposited per particle has increased. Ions, Mesons, Leptons and Baryons are byproducts of the Hadronic interactions which would mean that the energy deposited is distributed among these secondary particles. These particles are produced due to nuclear fragmentation, strong force interactions and through the decay of the secondary particles. It can also be observed that the total energy deposited has increased as well which would mean that the cells around the tumor region would undergo a significant impact.

EDP: Energy deposit per particle (MeV)

Table 2: Gamma beams: 50 (MeV)

| **Particles** | **EDP: Before Hadronic interactions** | **EDP: After Hadronic interactions** |
|---|---|---|
| Leptons | 1.000 | 0.9948 |
| Mesons | 0.000 | 0.0000 |
| Baryons | 0.000 | 0.0020 |
| ions | 0.0000 | 0.0031 |
| Total energy deposited | 10.9040 | 11.3440 |

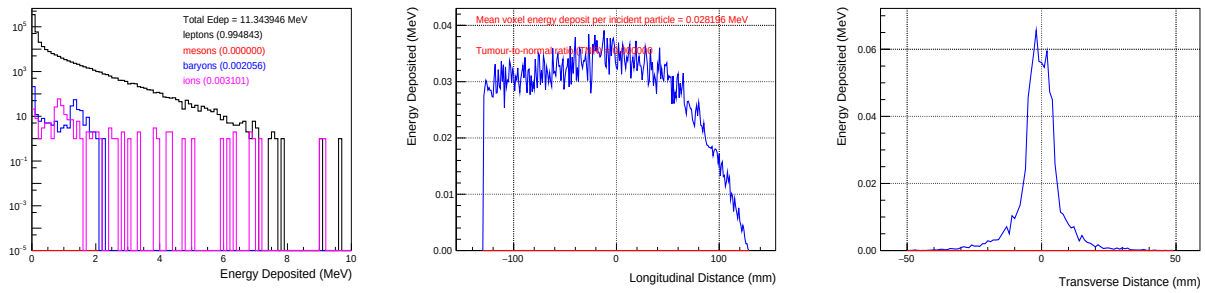
Table 3: Proton beams: 130 (MeV)

| Particles | EDP: Before Hadronic interactions | EDP: After Hadronic interactions |
|---|---|---|
| Leptons | 0.031 | 0.032 |
| Mesons | 0.000 | 0.000 |
| Baryons | 0.969 | 0.958 |
| ions | 0.000 | 0.097 |
| Total energy deposited | 129.9946 | 125.6270 |

Table 4: Carbon ion beams: 3000 (MeV)

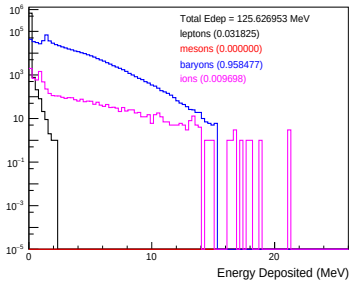
| Particles | EDP: Before Hadronic interactions | EDP: After Hadronic interactions |
|---|---|---|
| Leptons | 0.0587 | 0.05880 |
| Mesons | 0.0000 | 0.0000 |
| Baryons | 0.0000 | 0.0515 |
| ions | 0.9413 | 0.8897 |
| Total energy deposited | 2953.5344 | 2686.2268 |

Carbon ion beams produce more secondary particles in comparison. This is due to the fact that carbon beams are more heavier and possess a higher charge and are most likely to undergo nuclear fragmentation. Protons have a much smaller mass and therefore has less probability of nuclear fragmentation in comparison to carbon ions. The secondary particles created contains less energy. Finally photon beams have lowest impact when considering the creation of secondary particles as the primary mode of particle production is through Compton scattering and photoelectric absorption and they do not produce secondary particles as heavy ions do. It can be seen in Table 4 that the energy distribution is extremely similar and is not as significant as heavy ions. The secondary particles deposits energy out of the primary beam path mainly around the Bragg peak resulting in an increase in the dose in those regions. This might lead to unintended dose distributions around healthy tissues as well. The figures shown below are when the Hadronic interactions are introduced. As seen in Figure 17(c) the tail at the end of the energy deposition histogram is where a small amount of radiation that travels beyond the peak that can affect the healthy tissues.

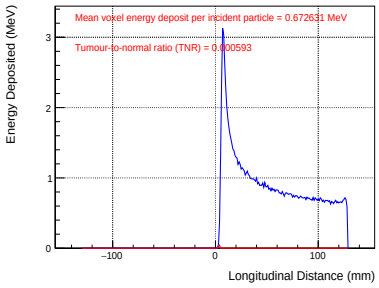



(a) Dose Profile (b) Longitudinal Dose Profile (c) Transverse Dose Profile

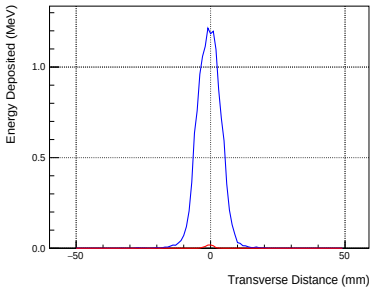
Figure 15: Gamma beam: 50 MeV




(a) Dose Profile

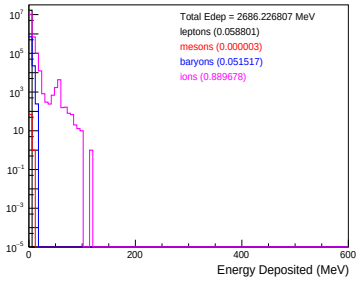



(b) Longitudinal Dose Profile

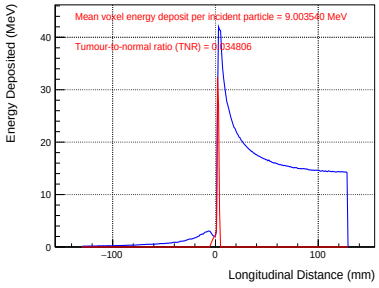



(c) Transverse Dose Profile

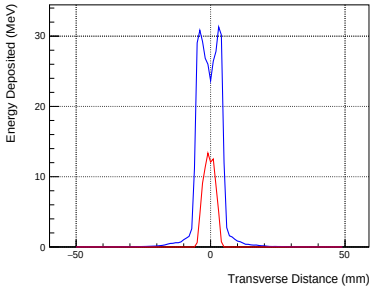
Figure 16: Proton beam: 130 MeV




(a) Dose Profile




(b) Longitudinal Dose Profile




(c) Transverse Dose Profile

Figure 17: Proton beam: 130 MeV

[Word Count: 995]

# 6 Conclusion

The simulation was done to evaluate the proton beams, photon beams and the carbon ion beams and its behavior when interacting with matter. It was observed that the primary understanding is to project the beam at the target area without causing much damage to the surrounding healthier tissues. According to the evaluations it can be concluded that carbon ions are better at radiation therapy due to the fact that they are able to offer the most accurate projection and higher biological effective due to linear energy transfer which allows better tumor destruction without causing much damage to the surrounding healthy tissues. Proton beams too are very precise and deposits the energy at the required target region but photon therapy is more applicable when the energy is lower. Proton beams can be used where high precision is required near sensitive organs. Photons are not exactly the best choice due to the fact that the beam is projected throughout the path causing damage to the entire region. In conclusion, more critical evaluations should be done to identify the most suitable beam depending on several parameters that involves the target region. According to this evaluation carbon ion beams seems more precise and accomplishes the required objective.
[Word Count: 203]

# 7 References

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